28/09/22

UNIT - I

$\cos iy = \cosh y$
$\sin iy = i \sinh y$

Function of complex variable

Continuity, derivative & analytic function - necessary condition for cauchy riemann equation theorem. sufficient condition (excluding proof) - Harmonic function - construction of analytic function

2 marks:

Define analytic function

A function $f(z)$ is said to be analytic at a point z_0 if f is differentiable not only at z_0 but at every point of neighbourhood.

$$z = x + iy$$
$$f(z) = u + iv$$
$$\bar{z} = x - iy$$

$f(z) = w$ same $w = u + iv$

Cauchy riemann equation: (CR equation) in cartesian coordinates. (or) necessary condition for $f(z)$ to be analytic (or) CR equation:

Statement:

If $f(z) = u(x,y) + iv(x,y)$ is an analytic function of z then the first order partial derivatives $\frac{\partial u}{\partial x}$, $\frac{\partial u}{\partial y}$, $\frac{\partial v}{\partial x}$ & $\frac{\partial v}{\partial y}$ should exist & satisfy

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \& \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

(or)

$$u_x = v_y \quad \& \quad u_y = -v_x$$

Proof:

Let $f(z) = u(x,y) + iv(x,y)$ be an analytic function then $f'(z)$ exist

$$f'(z) = \lim_{\Delta z \to 0} \frac{f(z+\Delta z) - f(z)}{\Delta z}$$

$$z = x + iy$$
$$\Delta z = \Delta x + i\Delta y$$

$$(z+\Delta z) = (x+\Delta x) + i(y+\Delta y)$$

$$f(z+\Delta z) = u(x+\Delta x, y+\Delta y) + iv(x+\Delta x, y+\Delta y)$$

$$f'(z) = \lim_{\substack{\Delta x \to 0 \\ \Delta y \to 0}} \left[\frac{u(x+\Delta x, y+\Delta y) + iv(x+\Delta x, y+\Delta y) - [u(x,y) + iv(x,y)]}{\Delta x + i\Delta y}\right]$$

$$= \lim_{\substack{\Delta x \to 0 \\ \Delta y \to 0}} \left[\frac{u(x+\Delta x, y+\Delta y) - u(x,y)}{\Delta x + i\Delta y}\right] + i \lim_{\substack{\Delta x \to 0 \\ \Delta y \to 0}} \left[\frac{v(x+\Delta x, y+\Delta y) - v(x,y)}{\Delta x + i\Delta y}\right]$$

Case 1:

when Δz is real

$$\Delta z = \Delta x \quad , \Delta y = 0$$

$$f'(z) = \lim_{\Delta x \to 0} \left[\frac{u(x+\Delta x, y) - u(x,y)}{\Delta x}\right] + i \lim_{\Delta x \to 0} \left[\frac{v(x+\Delta x, y) - v(x,y)}{\Delta x}\right]$$

$$\boxed{f'(z) = \frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x}} \quad \text{— (1)}$$

Case 2 :

When Δz is imaginary

$$\Delta z = \Delta y \quad \& \quad \Delta x = 0$$

$$f'(z) = \lim_{\Delta y \to 0}\left[\frac{u(x, y+\Delta y) - u(x,y)}{i\Delta y}\right] + \cancel{i}\lim_{\Delta y \to 0}\left[\frac{v(x, y+\Delta y) - v(x,y)}{\cancel{i}\Delta y}\right]$$

$$\left[\because \frac{1}{i} = -i\right]$$

$$= \lim_{\Delta y \to 0} -i\left[\frac{u(x, y+\Delta y) - u(x,y)}{\Delta y}\right] + \lim_{\Delta y \to 0}\left[\frac{v(x, y+\Delta y) - v(x,y)}{\Delta y}\right]$$

$$\boxed{f'(z) = -i\frac{\partial u}{\partial y} + \frac{\partial v}{\partial y}} \quad — (2)$$

From eqn (1) & (2)

compare real part

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad (\text{or}) \quad \boxed{u_x = v_y}$$

compare imaginary part

$$\frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y} \quad (\text{or}) \quad \boxed{v_x = -u_y}$$

Hence proved//

(*) 2m
Chauchy reimann equation in polar form :

Let (r, θ) be the polar coordinates whose cartesian coordinates of x, y is given by

$x = r\cos\theta$
$y = r\sin\theta$

then, $\frac{\partial u}{\partial r} = \frac{1}{r}\frac{\partial v}{\partial \theta}$ & $\frac{\partial v}{\partial r} = -\frac{1}{r}\frac{\partial u}{\partial \theta}$

Construction of analytic function (Real & imaginary

Milint Thomson method for real part:

$$f(z) = \int u_x(z,0) - i u_y(z,0)\, dz + c$$

Milint Thomson method for imaginary part:

$$f(z) = \int V_y(z,0) + i V_x(z,0)\, dz + c$$

construction of f(z) for real part

1. Find $f(z) = u + iv$, for the real part is $u = \log(x^2+y^2)$ & find v

<u>Soln:</u>

$$u = \log(x^2+y^2)$$

$$u_x = \frac{1}{x^2+y^2}(2x)$$

$$= \frac{2x}{x^2+y^2}$$

$$u_x(z,0) = \frac{2(z)}{z^2+0}$$

$$= \frac{2\cancel{z}}{z^{\cancel{2}}}$$

$$u_x(z,0) = \frac{2}{z}$$

$$u_y = \frac{1}{x^2+y^2}(2y)$$

$$= \frac{2y}{x^2+y^2}$$

$$U_y(z,0) = \frac{2(0)}{z^2+0^2} = 0$$

Using Milint Thomson method (real part)

$$f(z) = \int U_x(z,0) - i\,U_y(z,0)\,dz + c$$

$$= \int \left[\frac{2}{z} - 0\right] dz + c$$

$$= \int \frac{2}{z}\,dz + c$$

$$\boxed{f(z) = 2\log z + c}$$

To find v:

we use $f(z) = 2\log z$

$$u + iv = 2\log(x+iy)$$

$$\left[\because \log(a+ib) = \log\sqrt{a^2+b^2} + i\tan^{-1}\left(\frac{b}{a}\right)\right]$$

$$u + iv = 2\left[\log\sqrt{x^2+y^2} + i\tan^{-1}\left(\frac{y}{x}\right)\right]$$

$$= 2\left[\log(x^2+y^2)^{1/2} + i\tan^{-1}\left(\frac{y}{x}\right)\right]$$

$$= 2\left[\frac{1}{2}\log(x^2+y^2) + i\tan^{-1}\left(\frac{y}{x}\right)\right]$$

$$u + iv = \log(x^2+y^2) + 2i\tan^{-1}\left(\frac{y}{x}\right)$$

$$\boxed{v = 2\tan^{-1}\left(\frac{y}{x}\right)}$$

$d(uv) = uv' + vu'$

2) Find the analytic function whose real part is $e^{x}(x\cos y - y\sin y)$

Soln:

Gn:

$$u = e^{x}(x\cos y - y\sin y)$$

$$u_x = e^{x}[\cos y] + [x\cos y - y\sin y]e^{x}$$

$$u_x(z,0) = e^{z}(\cos 0) + [z(\cos 0) - 0]e^{z}$$

$$= e^{z} + ze^{z}$$

$$\boxed{u_x(z,0) = e^{z}(1+z)}$$

$$u_y = e^{x}[-x\sin y - (y\cos y + \sin y(1))] + [x\cos y - y\sin y]0$$

$$u_y(z,0) = e^{z}[0 - 0]$$

$$\boxed{u_y(z,0) = 0}$$

using milint thomson method (real part):

$$f(z) = \int u_x(z,0) - i\,u_y(z,0)\,dz + c$$

$$= \int [e^{z}(1+z) - 0]\,dz + c$$

$$= \int e^{z}(1+z)\,dz + c$$

$$= \int (e^z + ze^z)\,dz + c$$

$$= \cancel{e^z} + [ze^z - \cancel{e^z}] + c$$

$$\boxed{f(z) = ze^z + c}$$

$(1+z)e^z dz$

$u = 1+z \quad dv = e^z dz$

$u' = 1 \quad v = e^z$

$u'' = 0 \quad v_1 = e^z$

$(1+z)e^z - e^z$

$\cancel{e^z} + ze^z - \cancel{e^z}$

$$d\left(\frac{u}{v}\right) = \frac{vu' - uv'}{v^2}$$

3. Find the analytic function of $f(z)$. Given

$$u = \frac{\sin 2x}{\cosh 2y + \cos 2x}.$$

Soln:

$d(\cos 2\theta) = (-\sin 2\theta)2$

$d(\cosh 2\theta) = (\sinh 2\theta)2$

hyperbolic always +ve

$$Gn: u = \frac{\sin 2x}{\cosh 2y + \cos 2x}$$

$$u_x = \left[\frac{(\cosh 2y + \cos 2x)(2\cos 2x) - \sin 2x(-2\sin 2x)}{(\cosh 2y + \cos 2x)^2}\right]$$

$$= \left[\frac{2\cos 2x \cosh 2y + 2\cos^2 2x + 2\sin^2 2x}{(\cosh 2y + \cos 2x)^2}\right]$$

$$= 2\left[\frac{\cos 2x \cosh 2y + (\cos^2 2x + \sin^2 2x)}{(\cosh 2y + \cos 2x)^2}\right]$$

$$= 2\left[\frac{\cos 2x \cosh 2y + 1}{(\cosh 2y + \cos 2x)^2}\right]$$

$$u_x(z,0) = 2\left[\frac{\cos 2z \cosh 2y + 1}{(\cosh 2y + \cos 2z)^2}\right]$$

$$= 2\left[\frac{\cos 2z + 1}{(1+\cos 2z)^2}\right]$$

$$\boxed{u_x(z,0) = \frac{2}{1+\cos 2z}}$$

$$u_y = \left[\frac{(\cosh 2y + \cos 2x)(0) - \sin 2x (\sinh 2y) 2}{(\cosh 2y + \cos 2x)^2}\right]$$

$$= \frac{-2\sin 2x \sinh 2y}{(\cosh 2y + \cos 2x)^2}$$

$$u_y(z,0) = \frac{-2\sin 2z \sinh 2(0)}{(\cosh 2(0) + \cos 2z)^2}$$

$$= \frac{0}{(\cos 2z)^2}$$

$$\boxed{u_y(z,0) = 0}$$

Using Milint thomson method (real part)

$$f(z) = \int u_x(z,0) - i\, u_y(z,0)\, dz + c$$

$$= \int \left[\frac{2}{1+\cos 2z} - 0\right] dz + c$$

$$= \int \frac{2}{1+\cos 2z}\, dz + c$$

$$= 2 \int \frac{1}{1+\cos 2z}\, dz + c$$

$$\frac{1+\cos 2\theta}{2} = \cos^2\theta$$

$$\frac{1-\cos 2\theta}{2} = \sin^2\theta$$

$$= \int \frac{\cancel{2}}{\cancel{2}\cos^2 z}\, dz + c$$

$$d(\tan\theta) = \sec^2\theta$$

$$= \int \sec^2 z\, dz + c$$

$$d(\cot\theta) = -\operatorname{cosec}^2\theta$$

$$\cosh\theta = \frac{e^{\theta}+e^{-\theta}}{2}$$

$$\boxed{f(z) = \tan z + c}$$

$$\sinh\theta = \frac{e^{\theta}-e^{-\theta}}{2}$$

4. Find the analytic function for $f(z)$,

$$u = \frac{2\sin 2x}{e^{2y}+e^{-2y}-2\cos 2x}$$

<u>Soln:</u>

$$u = \frac{2\sin 2x}{2\cosh 2y - 2\cos 2x}$$

$$u = \frac{2\sin 2x}{2(\cosh 2y - \cos 2x)}$$

$$u = \frac{\sin 2x}{\cosh 2y - \cos 2x}$$

$$u_x = \left[\frac{(\cosh 2y - \cos 2x)(\cos 2x)2 - \sin 2x\,(\sin 2x)2}{(\cosh 2y - \cos 2x)^2}\right]$$

$$= \left[\frac{2\cos 2x \cosh 2y - 2\cos^2 2x - 2\sin^2 2x}{(\cosh 2y - \cos 2x)^2}\right]$$

$$= 2\left[\frac{\cos 2x \cosh 2y - (\cos^2 2x + \sin^2 2x)}{(\cosh 2y - \cos 2x)^2}\right]$$

$$= 2\left[\frac{\cos 2x \cosh 2y - 1}{(\cosh 2y - \cos 2x)^2}\right]$$

$$u_x(z,0) = 2\left[\frac{\cos 2z \cosh 2(0) - 1}{(\cosh 2(0) - \cos 2z)^2}\right]$$

$$= 2\left[\frac{\cos 2z - 1}{(1-\cos 2z)^2}\right]$$

$$= -2\left[\frac{-\cos 2z + 1}{(1-\cos 2z)^2}\right]$$

$$u_x(z,0) = \frac{-2}{1-\cos 2z}$$

$$u_y = \left[\frac{(\cosh 2y - \cos 2x)\,0 - \sin 2x\,(\sinh 2y)2}{(\cosh 2y - \cos 2x)^2}\right]$$

$$= \left[\frac{0 - 2\sin 2x \sinh 2y}{(\cosh 2y - \cos 2x)^2}\right]$$

$$u_y(z,0) = \left[\frac{-2\sin 2z \sinh 2(0)}{(\cosh 2(0) - (\cos 2z))^2}\right]$$

$$u_y(z,0) = 0$$

Using Milnt thomson method (real part)

$$f(z) = \int u_x(z,0) - i\,u_y(z,0)\,dz + c$$

$$= \int \frac{-2}{1-\cos 2z}\,dz + c$$

$$= -2\int \frac{1}{1-\cos 2z}\,dz + c$$

$$= -\cancel{2}\int \frac{1}{\cancel{2}\sin^2 z}\,dz + c$$

$$= -\int \operatorname{cosec}^2 z\,dz + c$$

$$f(z) = -(-\cot z) + c$$

$$\boxed{f(z) = \cot z + c}$$

Construction of Analytic function for imaginary Part

5. If $f(z) = u + iv$ & $v = \dfrac{2\sin x \sinh y}{\cos 2x + \cosh 2y}$

<u>Soln:</u>

Gn:

$$V = \frac{2\sin x \sinh y}{\cos 2x + \cosh 2y}$$

$$V_x = \left[\frac{(\cos 2x + \cosh 2y)(2\cos x \sinh y) - (2\sin x \sinh y)(-2\sin 2x)}{(\cos 2x + \cosh 2y)^2} \right]$$

$$V_x(z,0) = \left[\frac{(\cos 2z + 1)(0) - 0}{(\cos 2z + 1)^2} \right]$$

$$\boxed{V_x(z,0) = 0}$$

$$V_y = \left[\frac{(\cos 2x + \cosh 2y)(2\sin x \cosh y) - (2\sin x \sinh y)(\sinh 2y)2}{(\cos 2x + \cosh 2y)^2} \right]$$

$$V_y(z,0) = \left[\frac{\cancel{(\cos 2z + 1)}(2\sin z) - 0}{(\cos 2z + 1)^{\cancel{2}}} \right]$$

$$V_y(z,0) = \left[\frac{2\sin z}{(1 + \cos 2z)} \right]$$

Using Milint thomson method for imaginary part:

$$f(z) = \int V_y(z,0) + i V_x(z,0)\, dz + c$$

$$= \int \left[\frac{2 \sin 2z}{1+\cos 2z} + 0\right] dz + c$$

$d(\sec\theta) = \tan\theta \sec\theta$

$d(\operatorname{cosec}\theta) = -\cot\theta \operatorname{cosec}\theta$

$$= \int \frac{2 \sin 2z}{1+\cos 2z}\, dz + c$$

$$= \int \frac{\cancel{2} \sin 2z}{\cancel{2} \cos^2 z}\, dz + c$$

$$= \int \frac{\sin z}{\cos z} \cdot \frac{1}{\cos z}\, dz + c$$

$$= \int \tan z \sec z\, dz + c$$

$$\boxed{f(z) = \sec z + c}$$

6) Find the analytic function, given $v = e^{-2y}[y\cos 2x + x \sin 2x]$

<u>Soln</u>: $V = e^{-2y}[y\cos 2x + x\sin 2x]$

$$V_x = e^{-2y}[-2y\sin 2x + (2x\cos 2x + \sin 2x)] + [y\cos 2x + x\sin 2x](0)$$

$$V_x(z,0) = e^{-2(0)}[-2(0)\sin 2z + (2z\cos 2z + \sin 2z)]$$

$$\boxed{V_x(z,0) = 2z\cos 2z + \sin 2z}$$

$$V_y = e^{-2y}[\cos 2x] + [y\cos 2x + x\sin 2x](-2e^{-2y})$$

$$V_y(z,0) = e^{-2(0)}[\cos 2z] + [0 + z\sin 2z](-2e^{-2(0)})$$

$$\boxed{V_y(z,0) = \cos 2z - 2z\sin 2z}$$

Using Milint thomsom method for imaginary part:

$$f(z) = \int V_y(z,0) + i V_x(z,0)\, dz + c$$

$$= \int [(\cos 2z - 2z\sin 2z) + i(2z\cos 2z + \sin 2z)]\, dz + c$$

$$= \int [(\cos 2z + i\sin 2z) + (i2z\cos 2z - 2z\sin 2z)]\, dz + c$$

$$= \int e^{i2z} + [(i2z)(\cos 2z - \frac{1}{i}\sin 2z)]\, dz + c$$

$$= \int e^{i2z} + i2z(e^{i2z})\, dz + c \qquad [\because \frac{1}{i} = -i]$$

$$= \int e^{i2z}(1 + i2z)\, dz + c$$

$$= \int (1+i2z)e^{i2z}\, dz + c$$

$u = 1 + i2z$ | $dv = e^{i2z}$

$u' = 2i$ | $v = \dfrac{e^{i2z}}{2i}$

$v_1 = \dfrac{e^{i2z}}{(2i)^2}$

$$= \left[(1+i2z)\left(\frac{e^{i2z}}{2i}\right) - 2i\,\frac{e^{i2z}}{(2i)^2} \right] + c$$

$$= \left[(1+i2z)\left(\frac{e^{i2z}}{2i}\right) - \frac{e^{i2z}}{2i} \right] + c$$

$$= \frac{e^{i2z}}{2i}\left[(1+i2z) - 1\right] + c$$

$$\boxed{f(z) = \frac{e^{i2z}}{2i}\, i2z + c}$$

<u>2 marks:</u>

1. write the C-R equations in cartesian coordinates (or) write down C-R equations for an analytic function $f(z)$

$$f(z) = u(x,y) + iv(x,y)$$

$$u_x = v_y \quad \& \quad u_y = v_x$$

(or)

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \& \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

2. State C-R equation in polar coordinates

$$u_r = \frac{1}{r} v_\theta \quad \& \quad u_\theta = -r v_r$$

(or)

$$\frac{\partial u}{\partial r} = \frac{1}{r}\frac{\partial v}{\partial \theta} \quad \& \quad \frac{\partial u}{\partial \theta} = -r\frac{\partial v}{\partial r}$$

3. State any two properties of analytic functions.

An analytic function with constant real Part is also constant.

An analytic function of constant modulus is constant.

$$\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 0$$

4. $f(z) = xy - iy$ is continuous & where but not differentiable.

$$f(z) = xy - iy$$

$$u = xy$$

$$v = -y$$

$u_x = y$	$v_x = 0$
$u_y = x$	$v_y = -1$

$\therefore$ $f(z)$ is continuous & not differentiable

5. verify the function $w = \sin z$ is analytic or not.

$$w = f(z)$$

$$f(z) = \sin z = \sin(x+iy) \qquad [\because z = x+iy]$$

$[\because \sin(a+b) = \sin a \cos b + \cos a \sin b]$

$$f(z) = \sin x \cos iy + \cos x \sin iy$$

$$f(z) = \sin x \cosh y + \cos x \, i \sinh y \qquad \left[\because \cos iy = \cosh y, \; \sin iy = i \sinh y\right]$$

$$u = \sin x \cosh y$$

$$v = \cos x \sinh y$$

$$u_x = \cos x \cosh y \quad \Big| \quad v_x = -\sin x \sinh y$$

$$u_y = \sin x \sinh y \quad \Big| \quad v_y = \cos x \cosh y$$

$\therefore$ $u_x = v_y$

$u_y = -v_x$

$\therefore$ function is analytic

Problems based on $u-v$, $u+v$, $2u+3v$, ...

1. Construct an analytic function $f(z) = u+iv$, given $u+v = \dfrac{\sin 2x}{\cosh 2y - \cos 2x}$

Soln:

W.K.T

$$f(z) = u+iv \quad — ①$$

$u+v$

co-eff of $u = 1$ so $\Rightarrow 1 \times ①$

multiply $+v = -1$

$-v = -1$

with i

$$1 \times ① \Rightarrow f(z) = u+iv$$

v co-eff of $v = 1$ so $\Rightarrow ① \times (-1)$

$i^2 = -1$

$-i^2 = 1$

$$-i \times ① \Rightarrow -if(z) = -iu - i^2 v$$

$$= -iu + v$$

$$f(z) - if(z) = (u+v) + i(v-u)$$

$$f(z)[1-i] = (u+v) + i(v-u)$$

$$U = \frac{\sin 2x}{\cosh 2y - \cos 2x}$$

$$U_x = \left[\frac{(\cosh 2y - \cos 2x)(\cos 2x \cdot 2) - \sin 2x(-\sin 2x \cdot 2)}{(\cosh 2y - \cos 2x)^2}\right]$$

$$= \left[\frac{2\cos 2x \cosh 2y - 2\cos^2 2x + 2\sin^2 2x}{\cancel{2(\cos 2x \cosh 2y)} \; (\cosh 2y - \cos 2x)^2} \right]$$

$$= 2 \left[\frac{\cos 2x \cosh 2y - 1}{(\cosh 2y - \cos 2x)^2} \right]$$

$$U_x(z,0) = 2 \left[\frac{\cos 2z \cosh(0) - 1}{(\cosh(0) - \cos 2z)^2} \right]$$

$$= 2 \left[\frac{\cos 2z - 1}{(1 - \cos 2z)^2} \right]$$

$$= -2 \left[\frac{1 - \cos 2z}{(1 - \cos 2z)^2} \right]$$

$$\boxed{U_x(z,0) = \frac{-2}{(1 - \cos 2z)}}$$

$$U_y = \left[\frac{(\cosh 2y - \cos 2x)(0) - \sin 2x (\sinh 2y \, 2)}{(\cosh 2y - \cos 2x)^2} \right]$$

$$U_y(z,0) = \left[\frac{\cancel{\cosh}\ 0 - 2\sin 2z \cdot \sinh(0)}{(\cosh(0) - \cos 2(z))^2} \right]$$

$$\boxed{U_y(z,0) = 0}$$

Using Milne Thomson method for real part

$$f(z)(1-i) = \int [U_x(z,0) - iU_y(z,0)]\,dz + c$$

$$= \int \frac{-2}{1-\cos 2z}\,dz + c$$

$$= \int \frac{-2}{2\sin^2 z}\,dz + c$$

$$= -\int \frac{1}{\sin^2 z}\,dz + c$$

$$= -\int \operatorname{cosec}^2 z\,dz + c$$

$$f(z)(1-i) = -(-\cot z) + c = \cot z + c$$

$$f(z) = \frac{\cot z}{1-i}$$

$$= \frac{\cot z}{1-i} \times \frac{1+i}{1+i}$$

$(1+i)(1-i) = 1^2 - i^2$
$= 1+1$
$= 2$

$$f(z) = \frac{(1+i)\cot z}{2} + c$$

2. Construct an analytic function $f(z) = u + iv$, given $2u + 3v = e^x[\cos y - \sin y]$

Soln:

W.K.T

$$f(z) = u + iv \quad \text{--- (1)}$$

$2 \times (1) \Rightarrow 2f(z) = 2u + i2v$

$-3i \times (1) \Rightarrow -3if(z) = -3iu - 3i^2 v$

$= -3iu + 3v$

$$2f(z) - 3if(z) = 2u + 3v + i(2v - 3u)$$

$$f(z)[2-3i] = (2u+3v) + i(2v-3u)$$

$$U = e^{x}[\cos y - \sin y]$$

$$U_x = e^{x}[0] + [\cos y - \sin y]e^{x}$$

$$U_x(z,0) = [\cos 0 - \sin 0]e^{z}$$

$$= [1-0]e^{z}$$

$$= e^{z}$$

$$U_y(z,0) = e^{x}[-\sin y - \cos y] + [\cos y - \sin y]0$$

$$= e^{x}[-\sin y - \cos y]$$

$$U_y(z,0) = e^{z}[-\sin(0) - \cos(0)]$$

$$= e^{z}[0-1]$$

$$U_y(z,0) = -e^{z}$$

using Milint thomson method for real part

$$f(z)[2-3i] = \int [U_x(z,0) - iU_y(z,0)]\,dz + c$$

$$= \int [e^{z} - (-ie^{z})]\,dz + c$$

$$= \int [e^{z} + ie^{z}]\,dz + c$$

$$\cancel{= e^z + e^z}$$

$$= \int \left[e^z (1+i)\right] dz + c$$

$\cancel{u = (1+i)}$ $\qquad$ $\cancel{dv} =$

$$f(z)[2-3i] = e^z(1+i) + c$$

$$f(z) = \frac{e^z(1+i)}{(2-3i)}$$

$$= \frac{e^z(1+i)}{2-3i} \times \frac{2+3i}{2+3i}$$

$$= \frac{e^z(1+i)(2+3i)}{2^2 - (3i)^2}$$

$$2^2 - (3i)^2 = 4 + 9$$
$$= 13$$

$$= \frac{e^z(1+i)(2+3i)}{13} + c$$

$$= \frac{e^z(2+3i+2i+3i^2)}{13} + c$$

$$\boxed{i^2 = -1}$$

$$f(z) = \frac{e^z(-1+5i)}{13} + c$$

Harmonic functions

Definition:

A function u which satisfies laplace equation is called harmonic

i.e. $\boxed{u_{xx} + u_{yy} = 0}$

function.

1) Prove that the function $u = x^3 - 3xy^2 + 3x^2 - 3y^2 + 1$ is harmonic. Find the conjugate harmonic or (v)

Soln:

$u_x = 3x^2 - 3y^2 + 6x$ $\quad u_y = -6xy - 6y$

$u_{xx} = 6x + 6$ $\quad u_{yy} = -6x - 6$

$$u_{xx} + u_{yy} = 6x + 6 - 6x - 6 = 0$$

$\therefore$ It is Harmonic

$\therefore u_x(z, 0) = 3z^2 + 6z$

$u_y(z, 0) = 0$

Using Milint Thomson method for real part

$$f(z) = \int u_x(z,0) - i u_y(z,0) \, dz + c$$

$$= \int [3z^2 + 6z] \, dz + c$$

$$= 3\frac{z^3}{3} + \frac{6z^2}{2} + c$$

$$\boxed{f(z) = z^3 + 3z^2 + c}$$

Find v:

$$f(z) = z^3 + 3z^2$$

$$u + iv = (x+iy)^3 + 3(x+iy)^2$$

$$= x^3 + (iy)^3 + 3x(iy)^2 + 3x^2(iy) + 3[x^2 + (iy)^2 + 2x(iy)]$$

$$= x^3 - iy^3 - 3xy^2 + 3ix^2y + 3x^2 - 3y^2 + 6ixy$$

$$u + iv = (x^3 - 3xy^2 + 3x^2 - 3y^2) + i(-y^3 + 3x^2y + 6xy) + c$$

$$\boxed{v = 3x^2y - y^3 + 6xy}$$

2) Prove that $u = \dfrac{x}{x^2+y^2}$ is harmonic. Find v

Soln:

$$u = \frac{x}{x^2+y^2}$$

$$u_x = \left[\frac{(x^2+y^2)(1) - x(2x)}{(x^2+y^2)^2}\right]$$

$$= \left[\frac{x^2+y^2-2x^2}{(x^2+y^2)^2}\right]$$

$$= \left[\frac{-x^2+y^2}{(x^2+y^2)^2}\right] = -\cancel{\left[\frac{(x^2+y^2)}{(x^2+y^2)^2}\right]}$$

$$u_{xx} = \left[\frac{(x^2+y^2)^2(-2x) - (-x^2+y^2)\,2(+x^2+y^2)\,2x}{(x^2+y^2)^4} \right]$$

$$= \left[\frac{-2x(x^2+y^2)^2 - (-x^2+y^2)\,4x(x^2+y^2)}{(x^2+y^2)^4} \right]$$

$$= \left[\frac{-2x(x^2+y^2)^2 - [(4x^3+4xy^2)(-x^2+y^2)]}{(x^2+y^2)^4} \right]$$

$$= \left[\frac{-2x(x^4+2x^2y^2+y^4) - (-4x^5+\cancel{4x^3y^2}-\cancel{4x^3y^2}+4xy^4)}{(x^2+y^2)^4} \right]$$

$$= \left[\frac{-2x^5-4x^3y^2-2xy^4+4x^5-4xy^4}{(x^2+y^2)^4} \right]$$

$$u_{xx} = \left[\frac{2x^5-4x^3y^2-6xy^4}{(x^2+y^2)^4} \right]$$

$$u_y = \left[\frac{(x^2+y^2)(0) - x(2y)}{(x^2+y^2)^2} \right]$$

$$\boxed{u_{y\cancel{y}} = \frac{-2xy}{(x^2+y^2)^2}}$$

$$u_{yy} = \left[\frac{(x^2+y^2)^2(-2x) - (-2xy)\,2(x^2+y^2)(2y)}{(x^2+y^2)^4} \right]$$

$$= \left[\frac{-2x(x^4+2x^2y^2+y^4) - \left[-(4xy)(x^2+y^2)(2y)\right]}{(x^2+y^2)^4} \right]$$

$$= \left[\frac{-2x^5-4x^3y^2-2xy^4 - \left[-8xy^2(x^2+y^2)\right]}{(x^2+y^2)^4} \right]$$

$$= \left[\frac{-2x^5-4x^3y^2-2xy^4 - \left[-8x^3y^2-8xy^4\right]}{(x^2+y^2)^4} \right]$$

$$= \left[\frac{-2x^5-4x^3y^2-2xy^4+8x^3y^2+8xy^4}{(x^2+y^2)^4} \right]$$

$$u_{yy} = \left[\frac{-2x^5+4x^3y^2+6xy^4}{(x^2+y^2)^4} \right]$$

$$u_{xx}+u_{yy} = \left[\frac{\cancel{2x^5}-\cancel{4x^3y^2}-\cancel{6xy^4}-\cancel{2x^5}+\cancel{4x^3y^2}+\cancel{6xy^4}}{(x^2+y^2)^4} \right] = 0$$

$\therefore$ It is harmonic

$$\therefore u_x(z,0) = \frac{-z^2+0}{(z^2+0)^2} = \frac{-\cancel{z^2}}{z^{\cancel{4}2}} = \frac{-1}{z^2}$$

$$\boxed{u_x(z,0) = -(z^{-2})}$$

$$u_y(z,0) = \frac{-2z(0)}{(z^2+0)^2} = 0$$

Using Milne thomson method for real part

$$f(z) = \int u_x(z,0) - i u_y(z,0)\, dz + c$$

$$= \int (-z^{-2})\, dz + c$$

$$= -\int z^{-2}\, dz + c$$

$$= -\left[\frac{z^{-2+1}}{-2+1}\right] + c$$

$$= -\frac{z^{-1}}{-1} + c$$

$$= z^{-1} + c$$

$$\boxed{f(z) = \frac{1}{z} + c}$$

Find v :

$$f(z) = \frac{1}{z}$$

$$u + iv = \frac{1}{x+iy}$$

$$= \frac{1}{x+iy} \times \frac{x-iy}{x-iy}$$

$$= \frac{x-iy}{(x^2-(iy)^2)} = \frac{x-iy}{x^2+y^2}$$

$$u+iv = \frac{x}{x^2+y^2} - i\frac{y}{x^2+y^2}$$

By comparing

$$\boxed{v = \frac{-y}{x^2+y^2}}$$

IIM

If $f(z)$ is analytic function Prove that

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right) |f(z)|^2 = 4\,|f'(z)|^2$$

Soln:

$$z = x+iy$$

$$\bar{z} = x-iy$$

Add $\quad z+\bar{z} = 2x$

$$\boxed{x = \frac{z+\bar{z}}{2}}$$

Sub $\quad z-\bar{z} = 2iy$

$$\boxed{y = \frac{z-\bar{z}}{2i}}$$

$$\boxed{\frac{1}{i} = -i}$$

$$x = \frac{z+\bar{z}}{2} \qquad\qquad y = \frac{z-\bar{z}}{2i}$$

$$\frac{\partial x}{\partial z} = \frac{1+0}{2} = \frac{1}{2} \qquad\qquad \frac{\partial y}{\partial z} = \frac{1-0}{2i} = \frac{1}{2i} = -\frac{i}{2}$$

$$\frac{\partial x}{\partial \bar{z}} = \frac{0+1}{2} = \frac{1}{2} \qquad\qquad \frac{\partial y}{\partial \bar{z}} = \frac{0-1}{2i} = \frac{-1}{2i} = \frac{i}{2}$$

Find $\frac{\partial}{\partial z}$

$$\frac{\partial}{\partial z} = \frac{\partial}{\partial x}\left(\frac{\partial x}{\partial z}\right) + \frac{\partial}{\partial y}\left(\frac{\partial y}{\partial z}\right)$$

$$= \frac{\partial}{\partial x}\left(\frac{1}{2}\right) + \frac{\partial}{\partial y}\left(\frac{-i}{2}\right)$$

$$\boxed{\frac{\partial}{\partial z} = \frac{1}{2}\left[\frac{\partial}{\partial x} - \frac{i\,\partial}{\partial y}\right]} \quad — ①$$

$$\frac{\partial}{\partial \bar{z}} = \frac{\partial}{\partial x}\left(\frac{\partial x}{\partial \bar{z}}\right) + \frac{\partial}{\partial y}\left(\frac{\partial y}{\partial \bar{z}}\right)$$

$$= \frac{\partial}{\partial x}\left(\frac{1}{2}\right) + \frac{\partial}{\partial y}\left(\frac{i}{2}\right)$$

$$\boxed{\frac{\partial}{\partial \bar{z}} = \frac{1}{2}\left[\frac{\partial}{\partial x} + \frac{i\,\partial}{\partial y}\right]} \quad — ②$$

From ① & ②

$(a - b)$ $\quad$ $(a + b)$

$$\frac{\partial}{\partial z}\cdot\frac{\partial}{\partial \bar{z}} = \frac{1}{4}\left[\frac{\partial}{\partial x} - i\frac{\partial}{\partial y}\right]\left[\frac{\partial}{\partial x} + i\frac{\partial}{\partial y}\right]$$

$(a^2 - b^2)$

$$= \frac{1}{4}\left[\frac{\partial^2}{\partial x^2} - i^2\frac{\partial^2}{\partial y^2}\right]$$

$$= \frac{1}{4}\left[\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right]$$

$$\therefore \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} = 4\left(\frac{\partial}{\partial z}\cdot\frac{\partial}{\partial \bar{z}}\right)$$

multiply by $|f(z)|^2$

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right)|f(z)|^2 = 4\left(\frac{\partial}{\partial z}\cdot\frac{\partial}{\partial \bar{z}}\right)|f(z)|^2$$

WKT

$$|z|^2 = [z\cdot\bar{z}]$$

$$= 4 \frac{\partial}{\partial z} \cdot \frac{\partial}{\partial \bar{z}} \left[f(z) f(\bar{z}) \right]$$

$$= 4 \frac{\partial}{\partial z} \left[f(z) f'(\bar{z}) \right] \qquad \frac{\partial}{\partial \bar{z}}(f(\bar{z})) = f'(\bar{z})$$

$$= 4 \left[f'(z) f'(\bar{z}) \right] \qquad \frac{\partial}{\partial x}(f(x)) = f'(x)$$

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right) |f(z)|^2 = 4 \, |f'(z)|^2$$

Hence proved

2 marks :-

1. Show that $u = 3x^2y - y^3$ is harmonic.

Soln:

$$u = 3x^2y - y^3$$

$$u_x = 6xy$$

$$u_{xx} = 6y$$

$$u_y = 3x^2 - 3y^2$$

$$u_{yy} = 0 - 6y$$

$$\therefore \boxed{u_{xx} + u_{yy} = 0}$$

$$6y - 6y = 0$$

It is harmonic

2. $f(z) = \bar{z}$, check analytic or not.

Soln:

$$f(z) = \bar{z}$$

$$u + iv = x - iy$$

$u = x$

$v = -y$

$u_x = v_y$

$u_y = -v_x$

$u_x = 1$ | $v_x = 0$

$u_y = 0$ | $v_y = -1$

$u_x \neq v_y$

$\therefore$ It is not Analytic